

Part A – Theory (Short Questions)

Statistics, Probability & Linear Algebra — Student Score Analysis Project

Answer the following in 3–5 sentences each

Q1. Explain Mean, Median, and Mode with a real-life example.

Mean is the average value, found by summing all data points and dividing by the total count. **Median** is the middle value when data is arranged in ascending order, and **Mode** is the value that occurs most frequently in the dataset.

For example, if seven employees earn monthly salaries of ₹20k, ₹22k, ₹22k, ₹25k, ₹27k, ₹30k, and ₹45k, the Mean salary comes out to about ₹27.3k, the Median (the middle value) is ₹25k, and the Mode (the most repeated value) is ₹22k.

This example shows how a single high outlier (₹45k) pulls the Mean upward, while the Median and Mode stay closer to what a "typical" employee actually earns.

In practice, HR teams use the Mean for overall budgeting, the Median to describe typical pay levels, and the Mode to identify the most common salary bracket in the company.



Fig 1: Mean, Median and Mode marked on a real salary dataset

Q2. What is the difference between Variance and Standard Deviation?

Variance measures the average of the squared deviations of each data point from the mean, describing how spread out the data is. **Standard Deviation** is simply the square root of the variance, which brings the measurement back into the same unit as the original data.

Because variance is a squared value, it is often large and hard to interpret directly (for example, in "marks²"), whereas standard deviation is expressed in the same unit as the data ("marks"), making it far more intuitive to read.

Both describe the dispersion or spread of a dataset, but standard deviation is preferred in most real-world reporting because it can be compared directly with the mean.

For instance, if students' exam scores have a standard deviation of 10 marks, we can simply say scores typically vary by about 10 marks from the average — a statement that is much harder to interpret using variance (100 marks²) alone.



Fig 2: Variance and Standard Deviation formulas

Q3. Define Normal Distribution and give one practical use case.

Normal Distribution is a symmetric, bell-shaped probability distribution in which most data values cluster around the mean, and frequency gradually decreases as values move further away from the mean in either direction.

It is fully described by two parameters — the mean (μ) and standard deviation (σ) — and follows the empirical rule, which states that about 68%, 95%, and 99.7% of all data fall within 1, 2, and 3 standard deviations of the mean respectively.

Many natural and social phenomena, such as human height, blood pressure, and measurement errors, approximately follow this distribution.

A practical use case is quality control in manufacturing, where the normal distribution is used to set acceptable tolerance limits for product dimensions — items falling outside 2–3 standard deviations from the target value are flagged as defective.

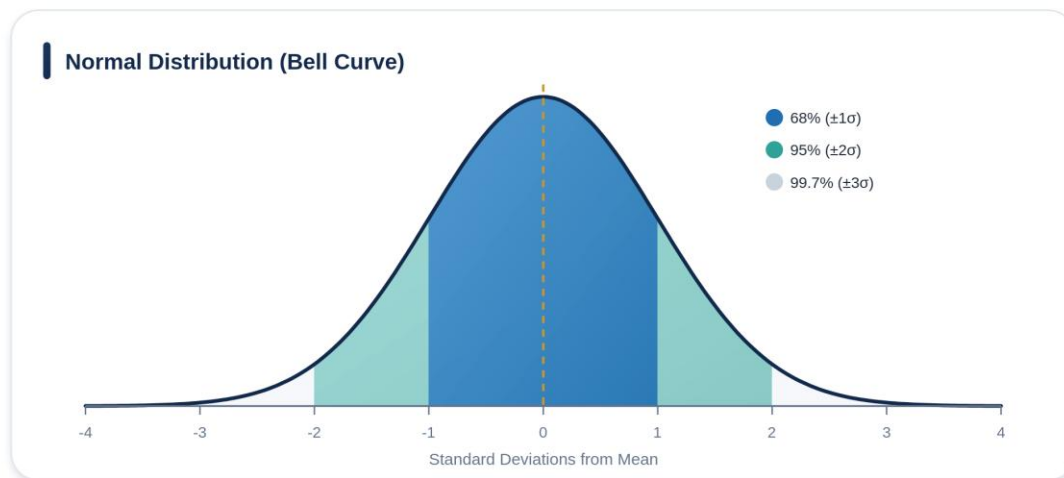


Fig 3: Normal distribution with 1σ , 2σ , and 3σ regions

Q4. Explain Skewness and Kurtosis in simple words.

Skewness measures the asymmetry of a data distribution — a **positive skew** means the tail is longer on the right side (most values are low with a few very high outliers), while a **negative skew** means the tail is longer on the left.

Kurtosis measures how "peaked" or "flat" a distribution is compared to a normal distribution — high kurtosis (leptokurtic) means a sharp peak with heavy tails, while low kurtosis (platykurtic) means a flatter shape with thinner tails.

Together, these two measures describe the shape of data beyond just its central tendency and spread, helping us understand how "typical" or "unusual" a distribution really looks.

For example, income data is usually right-skewed — most people earn near the average, but a small number earn extremely high incomes — which is why the mean income often appears higher than what most people actually earn.

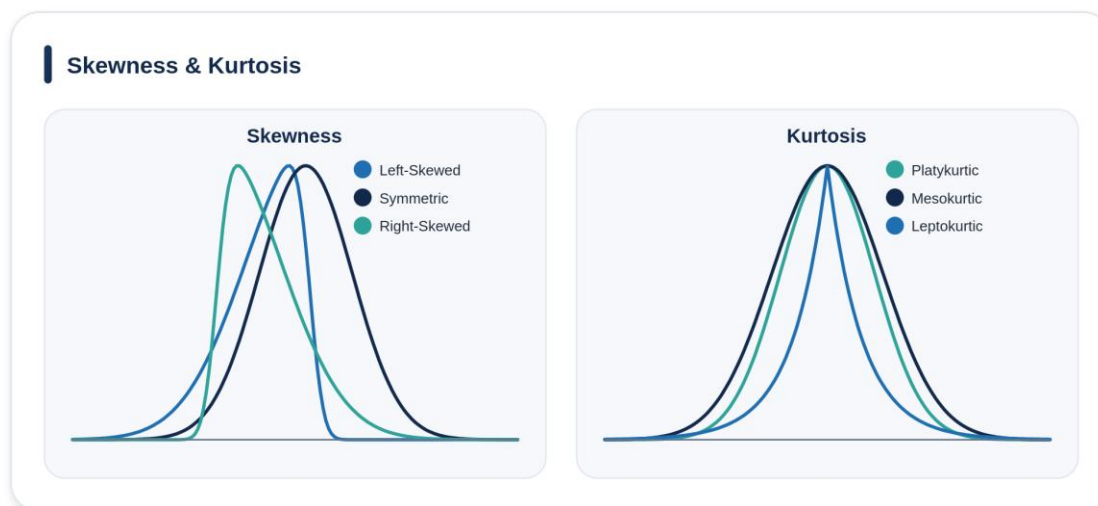


Fig 4: Skewness (left) and Kurtosis (right) shapes compared

Q5. What is Probability? Differentiate between Empirical vs Theoretical Probability with examples.

Probability is a numerical measure of how likely an event is to occur, ranging from 0 (impossible) to 1 (certain).

Theoretical Probability is calculated using possible outcomes with the formula $P(E) = n(E)/n(S)$, assuming all outcomes are equally likely — for example, the probability of rolling a 4 on a fair die is $1/6$.

Empirical Probability, on the other hand, is based on actual experiments or historical data, calculated as the number of times an event occurred divided by the total number of trials — for example, if a coin is tossed 100 times and lands on heads 47 times, the empirical probability of heads is $47/100 = 0.47$.

As the number of trials increases, empirical probability tends to converge toward theoretical probability — this idea is known as the Law of Large Numbers.



Fig 5: Theoretical vs Empirical Probability formulas

Q6. Explain Independent vs Dependent Events with one example each.

Independent events are events where the outcome of one does not affect the outcome of the other — for example, tossing a coin and rolling a dice are independent, since neither result influences the other.

Dependent events are events where the outcome of one does affect the probability of the other — for example, drawing two cards from a deck without replacement, since removing the first card changes the composition of the deck for the second draw.

Mathematically, independent events satisfy $P(A \cap B) = P(A) \times P(B)$, while dependent events follow $P(A \cap B) = P(A) \times P(B|A)$, where $P(B|A)$ is the conditional probability of B given A.

Correctly identifying whether events are independent or dependent is essential for accurately calculating combined probabilities in real-world situations such as insurance risk assessment or genetics.

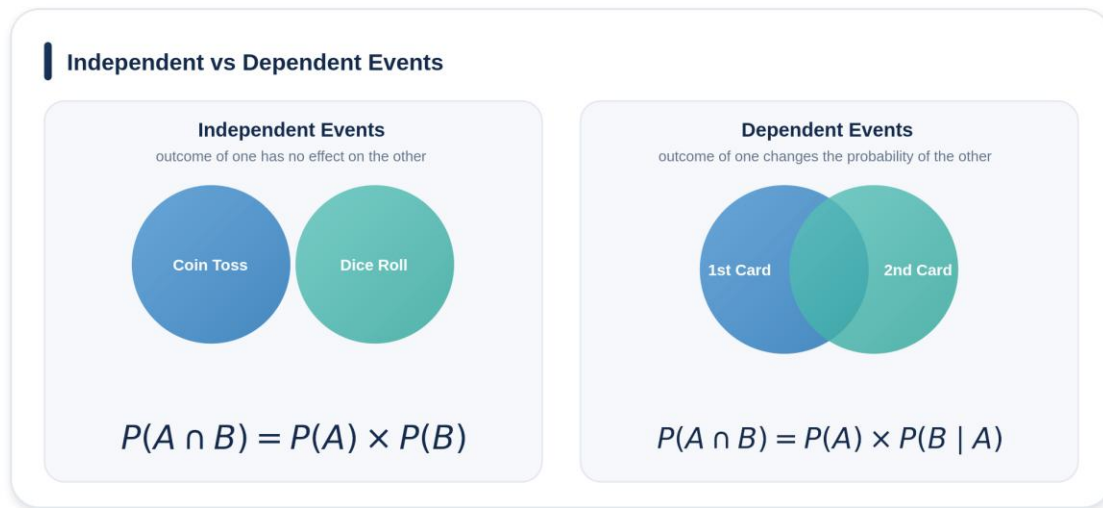


Fig 6: Independent events (no overlap) vs Dependent events (overlap)

Q7. What is the intuition of Bayes Theorem in daily life?

Bayes Theorem describes how to update the probability of a belief (hypothesis) when new evidence becomes available, using the formula $P(A|B) = [P(B|A) \times P(A)] / P(B)$.

In daily life, we do this intuitively — for instance, if you normally believe there is a low chance of rain (the prior belief), but then see dark clouds gathering (new evidence), you naturally revise your belief upward (the posterior belief) that it will rain.

The theorem simply formalizes this everyday process of "starting with an assumption, then refining it as new information arrives."

It is widely applied in real-world systems such as spam email filters, which continuously update the probability that a message is spam as new words or patterns are observed, and in medical diagnosis, where a positive test result updates the likelihood of disease based on the test's known accuracy.

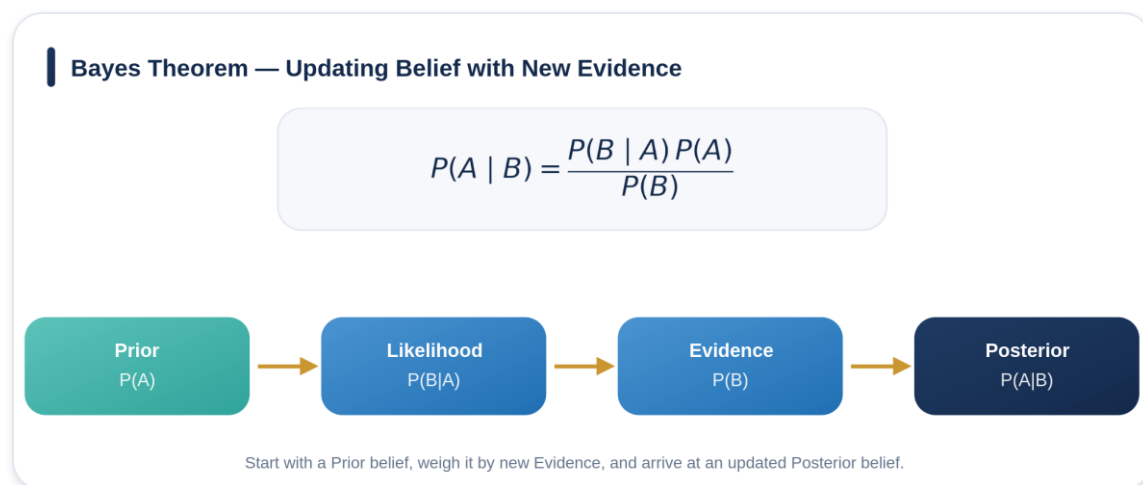


Fig 7: Bayes Theorem — updating belief with new evidence

Q8. Explain Eigenvalue and Eigenvector in simple terms.

Eigenvector of a matrix is a special non-zero vector that, when the matrix (representing a linear transformation) is applied to it, does not change direction — it is only scaled (stretched or shrunk).

Eigenvalue is the scalar factor by which the eigenvector is scaled, described by the equation $A \cdot v = \lambda v$, where A is the matrix, v is the eigenvector, and λ (lambda) is the eigenvalue.

In simple terms, eigenvectors represent the "natural directions" of a transformation, while eigenvalues tell us how much stretching or shrinking happens along each of those directions.

They are widely used in real-world applications such as facial recognition (Eigenfaces), Principal Component Analysis (PCA) for dimensionality reduction, and Google's PageRank algorithm for ranking web pages.

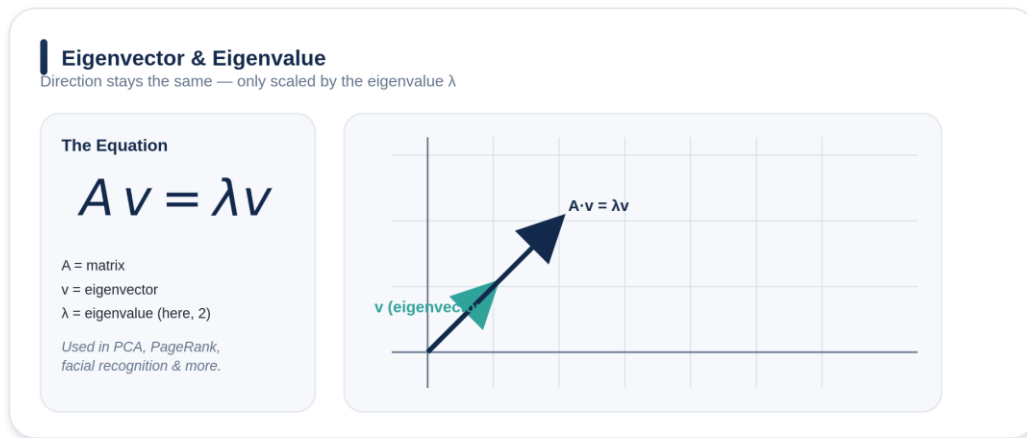


Fig 8: Eigenvector direction stays the same, only scaled by the eigenvalue λ